

Wireshark Attack Analysis Exercise

Victim Details:

- **IP Address:** 10.2.28.88
- **Host Name:** DESKTOP-TEYQ2NR
- **MAC Address:** 00:19:d1:b2:4d:ad
- **User Account Name:** brolf
- **Full Name of the user:** Becka Rolf

Malware type: NetSupport Manager RAT

2026-02-28-traffic-analysis-exercise.pcap

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb8a5cbb
2	1.476762	0.0.0.0	255.255.255.255	DHCP	379	DHCP Request - Transaction ID 0xaf98645
3	1.641635	Intel_b2:4d:ad	Broadcast	ARP	60	who has 10.2.28.1? Tell 10.2.28.88
4	1.641637	Cisco_d1:a2:25	Intel_b2:4d:ad	ARP	60	10.2.28.1 is at 00:05:00:11:a2:25
5	2.393884	Intel_b2:4d:ad	Broadcast	ARP	60	who has 10.2.28.2? Tell 10.2.28.88
6	2.393886	De11_2d:ce:69	Intel_b2:4d:ad	ARP	60	10.2.28.2 is at 14:b3:1f:2d:ce:69
7	2.473183	10.2.28.88	10.2.28.2	DNS	122	Standard query 0x7c4c SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net
8	2.473185	10.2.28.88	10.2.28.2	DNS	126	Standard query 0x23bd SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easys123.tech
9	2.473186	10.2.28.88	10.2.28.2	DNS	116	Standard query 0x62f5 SRV _ldap._tcp.Default-First-Site-Name._sites.easys123.tech
10	2.473483	10.2.28.2	10.2.28.88	DNS	189	Standard query response 0x23bd SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easys123.tech SRV 0 100 389 easys123.tech
11	2.473544	10.2.28.2	10.2.28.88	DNS	179	Standard query response 0x62f5 SRV _ldap._tcp.Default-First-Site-Name._sites.easys123.tech SRV 0 100 389 easys123.tech
12	2.473589	10.2.28.88	10.2.28.2	DNS	87	Standard query 0xbabd A easys123-dc.easys123.tech
13	2.473890	10.2.28.2	10.2.28.88	DNS	163	Standard query response 0xbabd A easys123-dc.easys123.tech A 10.2.28.2
14	2.474200	10.2.28.88	10.2.28.2	LDAP	265	searchRequest(97) "(<ROOT>)" baseObject
15	2.474201	10.2.28.88	10.2.28.2	LDAP	265	searchRequest(96) "(<ROOT>)" baseObject
16	2.474579	10.2.28.2	10.2.28.88	LDAP	229	searchResEntry(97) "(<ROOT>)" searchResDone(97) success [6 results]
17	2.474581	10.2.28.2	10.2.28.88	LDAP	229	searchResEntry(96) "(<ROOT>)" searchResDone(96) success [6 results]
18	2.567994	10.2.28.88	10.2.28.2	DNS	122	Standard query 0x7c4c SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.mshome.net
19	2.524461	10.2.28.88	10.2.28.2	DNS	126	Standard query 0xb5fd SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easys123.tech
20	2.524463	10.2.28.88	10.2.28.2	DNS	116	Standard query 0x0f9f SRV _ldap._tcp.Default-First-Site-Name._sites.easys123.tech
21	2.524662	10.2.28.2	10.2.28.88	DNS	189	Standard query response 0xb5fd SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.easys123.tech SRV 0 100 389 easys123.tech
22	2.524664	10.2.28.2	10.2.28.88	DNS	179	Standard query response 0x0f9f SRV _ldap._tcp.Default-First-Site-Name._sites.easys123.tech SRV 0 100 389 easys123-dc.ea

> Frame 1: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0.0.0.0
> Ethernet II, Src: RealtekSemi_68:08:00 (08:00:4c:68:08:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67
> Dynamic Host Configuration Protocol (Discover)

0000 ff ff ff ff ff ff 00 e0 4c 68 08 00 00 00 45 00LH...E:
0010 01 48 43 92 00 00 ff 11 77 13 00 00 00 00 ff ff ..HC...w...:
0020 ff ff 00 44 00 03 01 34 23 6f 01 01 00 00 a8 ..D.C.4.#...:
0030 5c bb 00 19 00 00 00 00 00 00 00 00 00 00 00 ..\.....Lh...:
0040 00 00 00 00 00 00 00 e0 4c 68 08 00 00 00 00LH...:
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:

2026-02-28-traffic-analysis-exercise.pcap Packets: 15512 Profile: Default

Wireshark - Endpoints - 2026-02-28-traffic-analysis-exercise.pcap

Endpoint Settings

☐ Name resolution
☐ Display raw data
☐ Hide aggregated
☐ Limit to display filter

Copy

Map

Protocol

☐ Bluetooth
☐ BPF7
☐ CCMP
☐ DNP 3.0
☒ Ethernet
☐ FC
☐ FDDI
☐ IEEE 802.11
☐ IEEE 802.15.4
☐ ILNP
☒ IPv4
☒ IPv6

Filter list for specific type

Address	Packets	Bytes	Tx Packets	Rx Packets	Tx Bytes	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
0.0.0.0	3	1 kB	3	1 kB	0	0 bytes						
4.149.160.182	38	18 kB	20	14 kB	18	4 kB						
10.2.28.1	1	349 bytes	1	349 bytes	0	0 bytes						
10.2.28.2	5,069	1 MB	2,326	571 kB	2,743	662 kB						
10.2.28.88	14,673	6 MB	7,405	2 MB	7,268	5 MB						
10.2.28.255	497	108 kB	0	0 bytes	497	108 kB						
13.69.116.109	47	20 kB	21	15 kB	26	5 kB						
13.70.79.200	30	9 kB	14	6 kB	16	4 kB						
13.89.178.27	29	9 kB	13	5 kB	16	4 kB						
13.89.179.8	24	10 kB	10	7 kB	14	2 kB						
13.89.179.9	28	9 kB	13	5 kB	15	4 kB						
13.89.179.13	45	20 kB	20	15 kB	25	5 kB						
13.89.179.14	35	12 kB	17	6 kB	18	6 kB						
13.107.213.57	73	23 kB	36	17 kB	37	6 kB						
13.107.246.57	31	11 kB	16	7 kB	15	4 kB						
20.42.65.84	68	25 kB	33	21 kB	35	4 kB						
20.42.65.88	69	25 kB	33	21 kB	36	4 kB						
20.42.65.94	78	35 kB	34	14 kB	44	21 kB						
20.42.73.30	28	9 kB	13	5 kB	15	3 kB						
20.48.190.241	24	7 kB	11	5 kB	13	2 kB						
20.72.205.209	24	7 kB	11	5 kB	13	2 kB						
20.96.153.111	32	12 kB	15	8 kB	17	4 kB						
20.106.86.13	77	25 kB	35	15 kB	42	10 kB						
20.189.173.1	46	20 kB	20	15 kB	26	5 kB						
20.189.173.2	44	16 kB	21	6 kB	23	10 kB						
20.189.173.8	31	9 kB	15	6 kB	16	4 kB						
20.189.173.14	34	12 kB	16	6 kB	18	6 kB						

Close Help

The endpoints confirm the victim IP is 10.2.28.88. The high packet count associated with the IP address indicates that the infected computer was 'checking-in' with the attacker's server thousands times for instructions. This also confirms that the malware was active and waiting for instruction.

2026-02-28-traffic-analysis-exercise.pcap

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No.	Time	Source	Destination	Protocol	Length	Info
100	4.219903	10.2.28.88	10.2.28.2	DNS	89	Standard query 0x2dc2 A v10.events.data.microsoft.com
101	4.257497	10.2.28.2	10.2.28.88	DNS	229	Standard query response 0x2dc2 A v10.events.data.microsoft.com CNAME win-global-asimov-leafs-events-data.trafficmanager.n
102	4.258596	10.2.28.88	13.89.179.9	TCP	66	59983 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
103	4.294204	0.0.0.0	255.255.255.255	DHCP	379	DHCP Request - Transaction ID 0xafaf98645
104	4.306185	10.2.28.1	10.2.28.88	DHCP	349	DHCP ACK - Transaction ID 0xafaf98645
105	4.321392	13.89.179.9	10.2.28.88	TCP	66	443 → 59983 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1376 WS=256 SACK_PERM
106	4.321590	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
107	4.324155	10.2.28.88	13.89.179.9	TLSv1.2	266	Client Hello (SH=V10.events.data.microsoft.com)
108	4.325517	10.2.28.88	224.0.0.252	LUMMR	75	Standard query 0x97f5 ANY DESKTOP-TEVQ2NR
109	4.386747	13.89.179.9	10.2.28.88	TCP	1410	443 → 59983 [ACK] Seq=1 Ack=213 Win=4195072 Len=1356 [TCP PDU reassembled in 112]
110	4.386959	13.89.179.9	10.2.28.88	TCP	1410	443 → 59983 [ACK] Seq=1357 Ack=213 Win=4195072 Len=1356 [TCP PDU reassembled in 112]
111	4.387173	13.89.179.9	10.2.28.88	TCP	1410	443 → 59983 [ACK] Seq=2713 Ack=213 Win=4195072 Len=1356 [TCP PDU reassembled in 112]
112	4.387174	13.89.179.9	10.2.28.88	TLSv1.2	221	Server Hello, Certificate, Server Key Exchange, Server Hello Done
113	4.387175	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=213 Ack=2713 Win=65280 Len=0
114	4.387405	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=213 Ack=4236 Win=65280 Len=0
115	4.393722	10.2.28.88	13.89.179.9	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
116	4.456382	13.89.179.9	10.2.28.88	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
117	4.456384	13.89.179.9	10.2.28.88	TLSv1.2	123	Application Data
118	4.456758	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=371 Ack=4356 Win=65280 Len=0
119	4.461435	10.2.28.88	13.89.179.9	TLSv1.2	141	Application Data
120	4.461437	10.2.28.88	13.89.179.9	TLSv1.2	962	Application Data
121	4.461438	10.2.28.88	13.89.179.9	TLSv1.2	92	Application Data
122	4.462296	10.2.28.88	13.89.179.9	TLSv1.2	1351	Application Data
123	4.521061	13.89.179.9	10.2.28.88	TLSv1.2	92	Application Data

Frame 105: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits)

Ethernet II, Src: Cisco_41:ae:25 (00:05:00:41:ae:25), Dst: Intel_b2:d1:b2:4d:ad (00:19:d1:b2:4d:ad)

Destination: Intel_b2:d1:b2:4d:ad (00:19:d1:b2:4d:ad)

Source: Cisco_41:ae:25 (00:05:00:41:ae:25)

Type: IPv4 (0x0800)

[Stream index: 2]

Internet Protocol Version 4, Src: 13.89.179.9, Dst: 10.2.28.88

Transmission Control Protocol, Src Port: 443, Dst Port: 59983, Seq: 0, Ack: 1, Len: 0

0000 00 19 d1 b2 4d ad 00 05 00 41 ae 25 00 00 45 28 ...M...A...E(
0010 00 34 9a 6b 40 00 0b 06 8e 74 bd 59 b3 09 0a 02 ...4-k...t.Y...
0020 1c 58 01 bb ea 4f 9f bc 25 a1 27 17 93 79 80 12 ...X...-...%...y...
0030 ff ff 1c 9f 00 00 02 04 05 60 01 03 03 08 01 01
0040 04 02

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No.	Time	Source	Destination	Protocol	Length	Info
100	4.219903	10.2.28.88	10.2.28.2	DNS	89	Standard query 0x2dc2 A v10.events.data.microsoft.com
101	4.257497	10.2.28.2	10.2.28.88	DNS	229	Standard query response 0x2dc2 A v10.events.data.microsoft.com CNAME win-global-asimov-leafs-events-data.trafficmanager.n
102	4.258596	10.2.28.88	13.89.179.9	TCP	66	59983 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
103	4.294204	0.0.0.0	255.255.255.255	DHCP	379	DHCP Request - Transaction ID 0xafaf98645
104	4.306185	10.2.28.1	10.2.28.88	DHCP	349	DHCP ACK - Transaction ID 0xafaf98645
105	4.321392	13.89.179.9	10.2.28.88	TCP	66	443 → 59983 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1376 WS=256 SACK_PERM
106	4.321590	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
107	4.324155	10.2.28.88	13.89.179.9	TLSv1.2	266	Client Hello (SH=V10.events.data.microsoft.com)
108	4.325517	10.2.28.88	224.0.0.252	LUMMR	75	Standard query 0x97f5 ANY DESKTOP-TEVQ2NR
109	4.386747	13.89.179.9	10.2.28.88	TCP	1410	443 → 59983 [ACK] Seq=1 Ack=213 Win=4195072 Len=1356 [TCP PDU reassembled in 112]
110	4.386959	13.89.179.9	10.2.28.88	TCP	1410	443 → 59983 [ACK] Seq=1357 Ack=213 Win=4195072 Len=1356 [TCP PDU reassembled in 112]
111	4.387173	13.89.179.9	10.2.28.88	TCP	1410	443 → 59983 [ACK] Seq=2713 Ack=213 Win=4195072 Len=1356 [TCP PDU reassembled in 112]
112	4.387174	13.89.179.9	10.2.28.88	TLSv1.2	221	Server Hello, Certificate, Server Key Exchange, Server Hello Done
113	4.387175	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=213 Ack=2713 Win=65280 Len=0
114	4.387405	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=213 Ack=4236 Win=65280 Len=0
115	4.393722	10.2.28.88	13.89.179.9	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
116	4.456382	13.89.179.9	10.2.28.88	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
117	4.456384	13.89.179.9	10.2.28.88	TLSv1.2	123	Application Data
118	4.456758	10.2.28.88	13.89.179.9	TCP	60	59983 → 443 [ACK] Seq=371 Ack=4356 Win=65280 Len=0
119	4.461435	10.2.28.88	13.89.179.9	TLSv1.2	141	Application Data
120	4.461437	10.2.28.88	13.89.179.9	TLSv1.2	962	Application Data
121	4.461438	10.2.28.88	13.89.179.9	TLSv1.2	92	Application Data
122	4.462296	10.2.28.88	13.89.179.9	TLSv1.2	1351	Application Data
123	4.521061	13.89.179.9	10.2.28.88	TLSv1.2	92	Application Data

Frame 103: Packet, 379 bytes on wire (3032 bits), 379 bytes captured (3032 bits)

Ethernet II, Src: Intel_b2:4d:ad (00:19:d1:b2:4d:ad), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: Intel_b2:4d:ad (00:19:d1:b2:4d:ad)

Type: IPv4 (0x0800)

[Stream index: 1]

Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255

User Datagram Protocol, Src Port: 68, Dst Port: 67

Dynamic Host Configuration Protocol (Request)

Message type: Boot Request (1)

Hardware type: Ethernet (0x01)

Hardware address: Intel_b2:4d:ad (00:19:d1:b2:4d:ad)

0000 ff ff ff ff ff ff 00 19 d1 b2 4d ad 08 00 45 00M...E:
0010 01 6d 12 44 00 00 80 11 27 3d 00 00 00 ff ff ...m-D...Y...
0020 ff ff 00 44 00 43 01 59 6d 34 81 01 00 00 af a9 ...-D-C-Y...
0030 86 45 00 03 00 00 00 00 00 00 00 00 00 00 00 ...E.....
0040 00 00 00 00 00 00 19 d1 b2 4d ad 00 00 00 00M...E:
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0120 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0130 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0140 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0190 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
01f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0200 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0210 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0220 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0230 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0240 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0250 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0260 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0270 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0280 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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02e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
02f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0300 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0310 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0320 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0330 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0340 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0350 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0360 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0370 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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03d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
03e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
03f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0400 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0410 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0420 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0430 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0440 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0450 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0460 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0470 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0480 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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04d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
04e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
04f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0500 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0510 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0520 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0530 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0540 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0550 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0560 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0570 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0580 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0590 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
05a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
05b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
05c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
05d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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0610 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0620 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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0660 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0670 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0680 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0690 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
06a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
06b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
06c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
06d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
06e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
06f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0700 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0710 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0720 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0730 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0740 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0750 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0760 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0770 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0780 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0790 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
07a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
07b0 00 00 00 00 0

2026-02-28-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xba85cbb
2	1.476762	0.0.0.0	255.255.255.255	DHCP	379	DHCP Request - Transaction ID 0xaf98645
103	4.294204	0.0.0.0	255.255.255.255	DHCP	379	DHCP Request - Transaction ID 0xaf98645
104	4.306185	10.2.28.1	10.2.28.88	DHCP	349	DHCP ACK - Transaction ID 0xaf98645

> Frame 103: Packet, 379 bytes on wire (3032 bits), 379 bytes captured (3032 bits)

Ethernet II, Src: Intel_b2:4d:ad (00:19:d1:b2:4d:ad), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255

User Datagram Protocol, Src Port: 68, Dst Port: 67

Dynamic Host Configuration Protocol (Request)

Message type: Boot Request (1)

Hardware type: Ethernet (0x01)

Hardware address length: 6

Hops: 0

Transaction ID: 0xaf98645

Seconds elapsed: 3

Bootp flags: 0x0000 (Unicast)

Client IP address: 0.0.0.0

Your (client) IP address: 0.0.0.0

Next server IP address: 0.0.0.0

Relay agent IP address: 0.0.0.0

Client MAC address: Intel_b2:4d:ad (00:19:d1:b2:4d:ad)

Client hardware address padding: 00000000000000000000

Server host name not given

Boot file name not given

Magic cookie: DHCP

> Option: (53) DHCP Message Type (Request)

> Option: (61) Client identifier

> Option: (50) Requested IP Address (10.2.28.88)

> Option: (12) Host Name

Length: 15

Host Name: DESKTOP-TEYQ2NR

> Option: (81) Client Fully Qualified Domain Name

> Option: (60) Vendor class identifier

> Option: (55) Parameter Request List

> Option: (255) End

Dynamic Host Configuration Protocol: Protocol

Packets: 15512 - Displayed: 4 (0.0%)

Profile: Default

Then after using the filter dhcp, we found the dhcp request (packet 103), where option: (12) mentions the hostname which is DESKTOP-TEYQ2NR.

2026-02-28-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Kerberos.CNameString

No.	Time	Source	Destination	Protocol	Length	Info
243	16.565177	10.2.28.88	10.2.28.2	KRB5	270	AS-REQ
251	16.570763	10.2.28.88	10.2.28.2	KRB5	350	AS-REQ
253	16.572761	10.2.28.2	10.2.28.88	KRB5	258	AS-REP
265	16.576639	10.2.28.2	10.2.28.88	KRB5	199	TGS-REP
288	16.745555	10.2.28.2	10.2.28.88	KRB5	333	TGS-REP
300	16.747703	10.2.28.2	10.2.28.88	KRB5	154	TGS-REP
369	16.813165	10.2.28.2	10.2.28.88	KRB5	303	TGS-REP
400	16.848884	10.2.28.2	10.2.28.88	KRB5	349	TGS-REP
2672	52.079058	10.2.28.2	10.2.28.88	KRB5	333	TGS-REP

Urgent Pointer: 0

> [Timestamps]

> [SEQ/ACK analysis]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

TCP payload (296 bytes)

[PDU Size: 296]

Kerberos

Record Mark: 292 bytes

as-req

pvno: 5

msg-type: krb-as-req (10)

padat: 2 items

req-body

Padding: 0

kdc-options: 0810010

cname

name-type: KRB5-NT-PRINCIPAL (1)

cname-string: 1 item

CNameString: broflf

realm: EASYAS123

sname

till: Sep 12, 2100 22:48:05.000000000 Eastern Daylight Time

rtime: Sep 12, 2100 22:48:05.000000000 Eastern Daylight Time

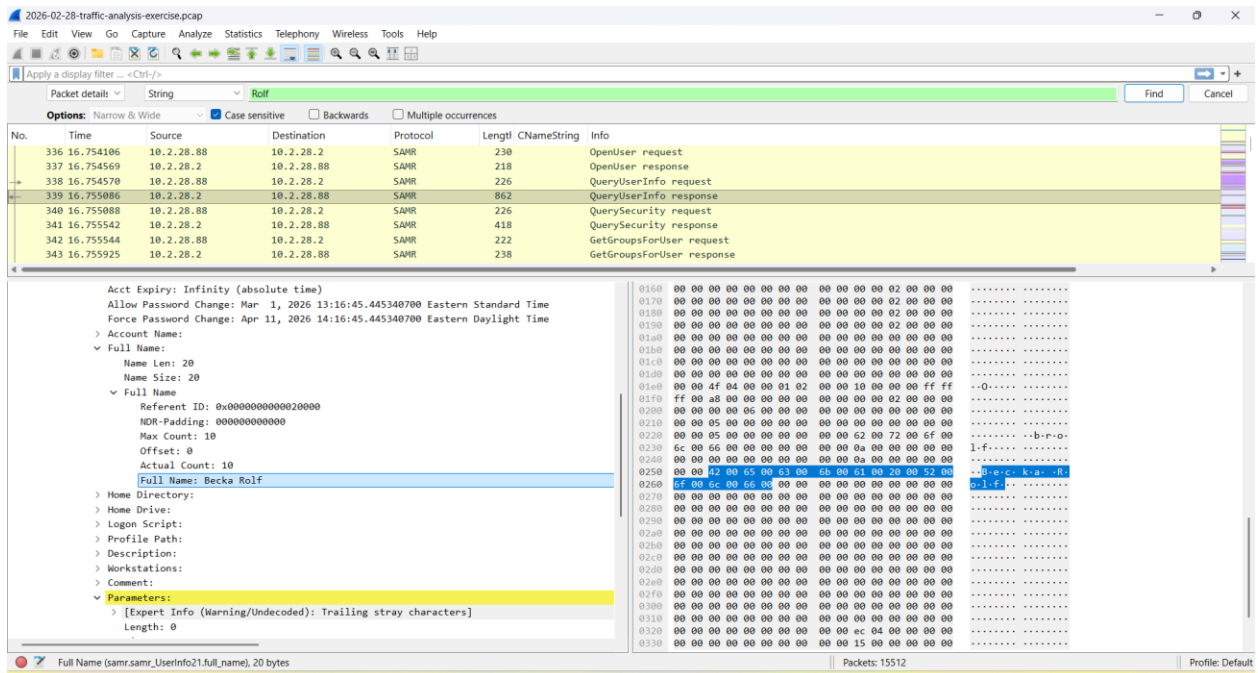
nonce: 296313762

2026-02-28-traffic-analysis-exercise.pcap

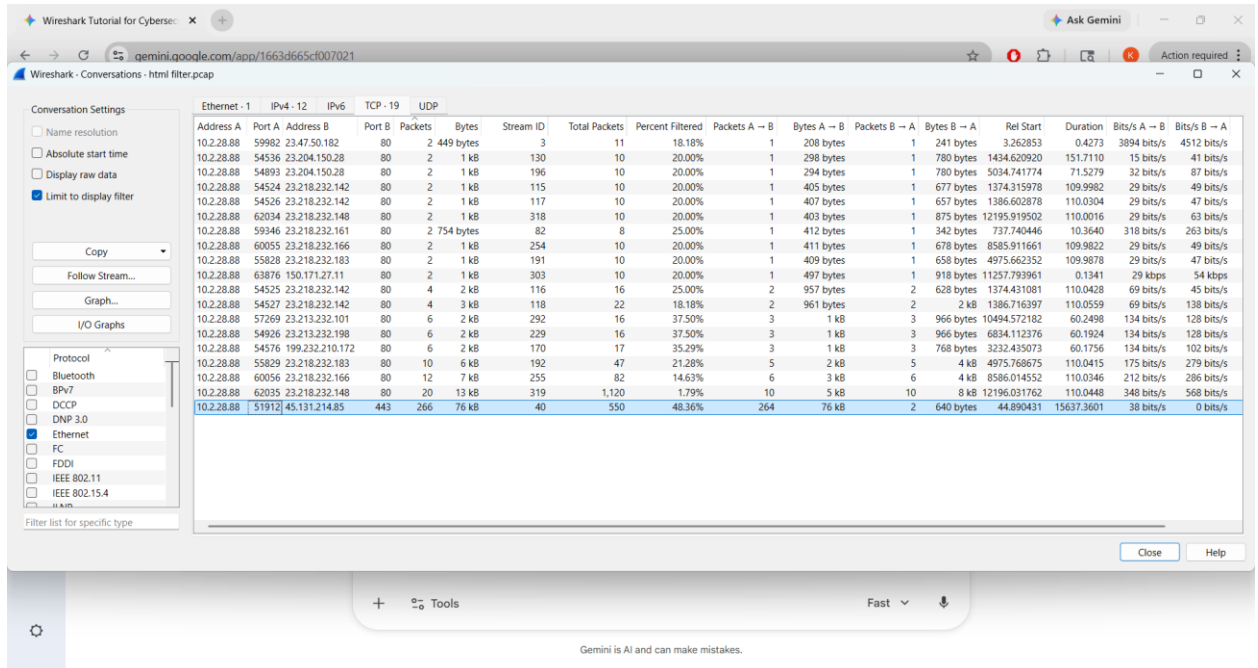
Packets: 15512 - Displayed: 9 (0.1%)

Profile: Default

By using the filter Kerberos.CNameString, we found out 2 AS-REQ packets, going in further detail by opening each packet, we found out the Username which is broflf.



We got the Username brof, with the help of that we did a search using Rolf as the keyword and we found the packet containing the full name as Becka Rolf.



The victim is 10.2.28.88 and the attacker is 45.131.214.85 which is using port 443 (https) for the communication. The duration of the connection shows 15637.3601 seconds which confirms active Command and Control channel.

2026-02-28-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.user_agent contains "NetSupport" or ip.addr == 45.131.214.85

No.	Time	Source	Destination	Protocol	Length	CNameString	Info
2569	44.898431	10.2.28.88	45.131.214.85	TCP	66		51912 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2636	45.040776	45.131.214.85	10.2.28.88	TCP	66		443 → 51912 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1376 WS=256 SACK_PERM
2637	45.040779	10.2.28.88	45.131.214.85	TCP	60		51912 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2638	45.048059	10.2.28.88	45.131.214.85	HTTP	272		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2639	45.195238	45.131.214.85	10.2.28.88	HTTP	278		HTTP/1.1 200 OK (application/x-www-form-urlencoded)
2640	45.196891	10.2.28.88	45.131.214.85	HTTP	494		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2641	45.355474	45.131.214.85	10.2.28.88	HTTP	362		HTTP/1.1 200 OK (application/x-www-form-urlencoded)
2642	45.404166	10.2.28.88	45.131.214.85	TCP	60		51912 → 443 [ACK] Seq=659 Ack=533 Win=64768 Len=0

[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (224 bytes)

Hypertext Transfer Protocol

- [Expert Info (Warning/Security): Unencrypted HTTP protocol detected over encrypted port, could indicate a dangerous misconfiguration
[Severity level: Warning]
[Group: Security]
- HTTP/1.1 200 OK
Response Version: HTTP/1.1
Status Code: 200
[Status Code Description: OK]
Response Phrase: OK
Server: NetSupport Gateway/1.92 (Windows NT)\nContent-Type: application/x-www-form-urlencoded\nContent-Length: 69\n[Content length: 69]
Connection: Keep-Alive\n\n[Request in frame: 2638]
[Time since request: 147.179000 milliseconds]
[Request URI: http://45.131.214.85/fakeurl.htm]
[Full request URI: http://45.131.214.85/fakeurl.htm]
File Data: 69 bytes
- HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "CMD" = "ENCOD(nES=1nDATA=eg \$%(\024% \005\sa%w*([R*?])023*\001d8*="s\177\030\002

2026-02-28-traffic-analysis-exercise.pcap Packets: 15512 - Displayed: 550 (3.5%) Profile: Default

2026-02-28-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.user_agent contains "NetSupport" or ip.addr == 45.131.214.85

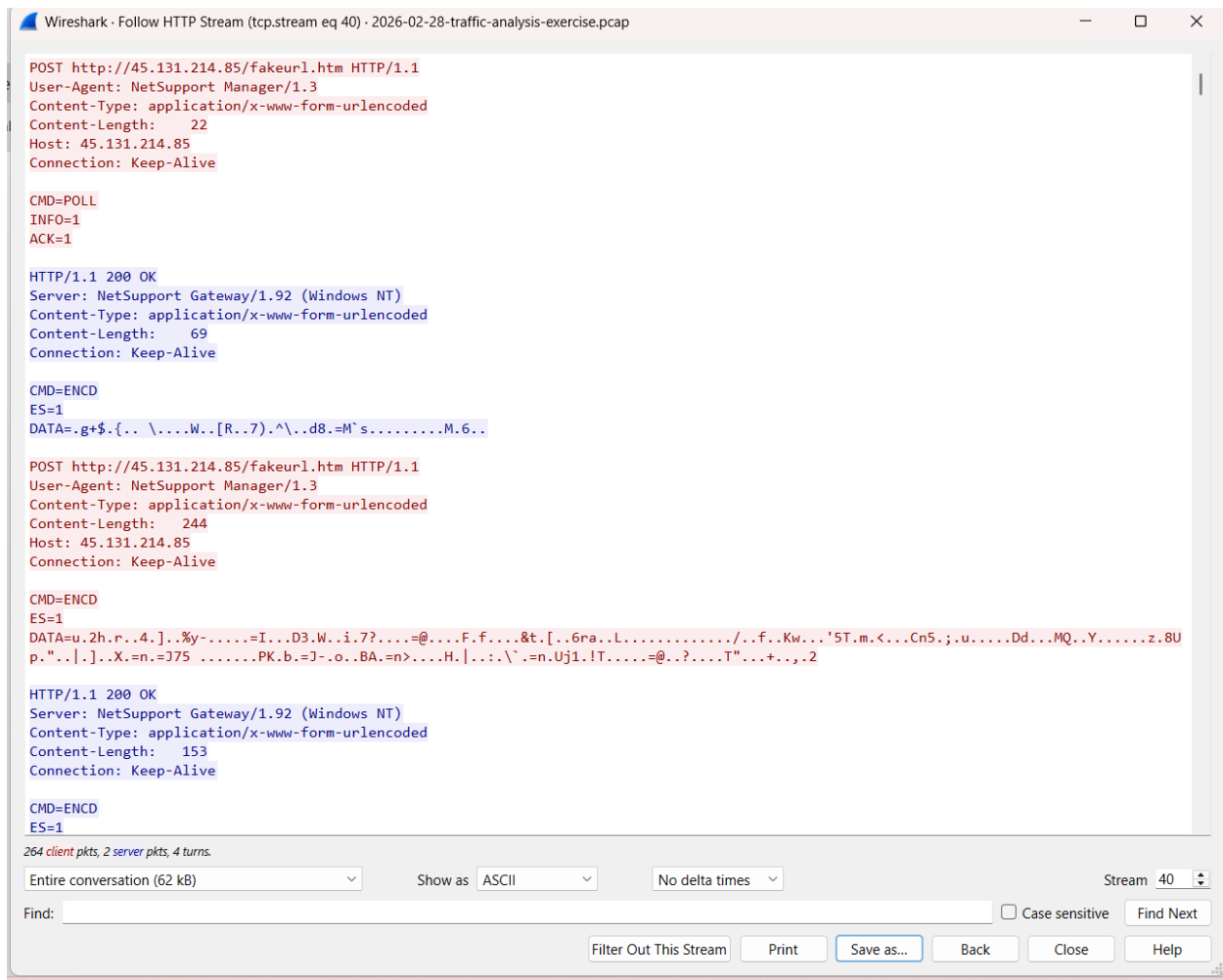
No.	Time	Source	Destination	Protocol	Length	CNameString	Info
2569	44.898431	10.2.28.88	45.131.214.85	TCP	66		51912 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2636	45.040776	45.131.214.85	10.2.28.88	TCP	66		443 → 51912 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1376 WS=256 SACK_PERM
2637	45.040779	10.2.28.88	45.131.214.85	TCP	60		51912 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2638	45.048059	10.2.28.88	45.131.214.85	HTTP	272		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2639	45.195238	45.131.214.85	10.2.28.88	HTTP	278		HTTP/1.1 200 OK (application/x-www-form-urlencoded)
2640	45.196891	10.2.28.88	45.131.214.85	HTTP	494		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2641	45.355474	45.131.214.85	10.2.28.88	HTTP	362		HTTP/1.1 200 OK (application/x-www-form-urlencoded)
2642	45.404166	10.2.28.88	45.131.214.85	TCP	60		51912 → 443 [ACK] Seq=659 Ack=533 Win=64768 Len=0
2643	45.555924	10.2.28.88	45.131.214.85	HTTP	326		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2644	45.754327	45.131.214.85	10.2.28.88	TCP	60		443 → 51912 [ACK] Seq=533 Ack=931 Win=2096640 Len=0
2645	45.756275	45.131.214.85	10.2.28.88	HTTP	327		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
2646	45.948862	45.131.214.85	10.2.28.88	TCP	60		443 → 51912 [ACK] Seq=533 Ack=1204 Win=2096384 Len=0
3807	105.932070	10.2.28.88	45.131.214.85	HTTP	286		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
3808	106.134592	45.131.214.85	10.2.28.88	TCP	60		443 → 51912 [ACK] Seq=533 Ack=1436 Win=2097664 Len=0
4093	166.005869	10.2.28.88	45.131.214.85	HTTP	286		POST http://45.131.214.85/fakeurl.htm HTTP/1.1 (application/x-www-form-urlencoded)
4094	166.208505	45.131.214.85	10.2.28.88	TCP	60		443 → 51912 [ACK] Seq=533 Ack=1668 Win=2097408 Len=0

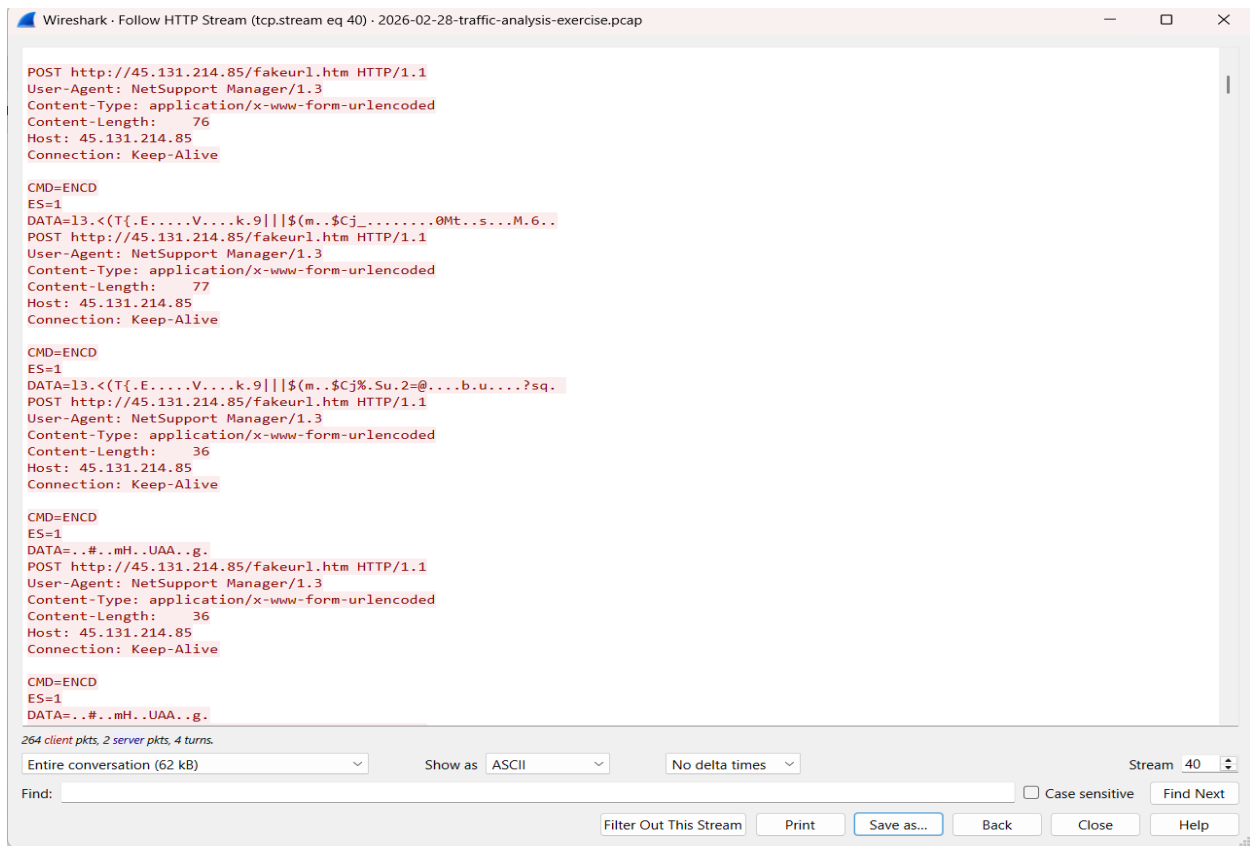
Response Version: HTTP/1.1
Status Code: 200
[Status Code Description: OK]
Response Phrase: OK
Server: NetSupport Gateway/1.92 (Windows NT)\nContent-Type: application/x-www-form-urlencoded\nContent-Length: 153\n[Content length: 153]
Connection: Keep-Alive\n\n[Request in frame: 2640]
[Time since request: 158.583000 milliseconds]
[Request URI: http://45.131.214.85/fakeurl.htm]
[Full request URI: http://45.131.214.85/fakeurl.htm]
File Data: 153 bytes

HTML Form URL Encoded: application/x-www-form-urlencoded

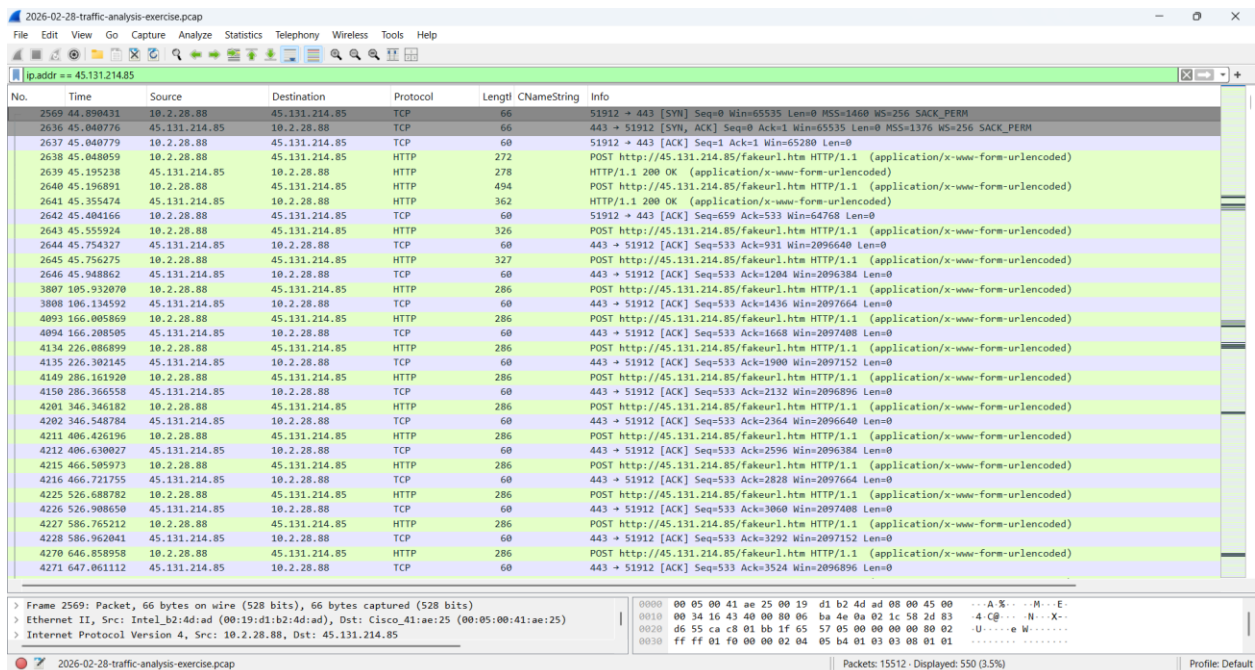
- Form item: "CMD" = "ENCOD(nES=1nDATA=2h\fr*\002 \005\sa%w*([R*?])023*\001d8*="s\177\030\002
- Form item: "\037\001\003" = "n*****"
- Form item: "Z*" = "n*\fr*Lq*f3*\004n*****\002*\005V*\177*('*\022p\003y*d*([R*?])022

2026-02-28-traffic-analysis-exercise.pcap Packets: 15512 - Displayed: 550 (3.5%) Profile: Default





While the initial delivery vector was not present in the capture, the infection was confirmed through behavioural analysis of 10.2.28.88. The host demonstrated immediate C2 beaoning patterns upon joining the network, indicating established persistence by NetSupport RAT.



This proves that no other machine was infected except 10.2.28.88.